Math Practice Test — Measuring detectives

Year 3 · AV3 · 3rd term (practice)

Name: _____ Date: _____ Class: _____ Score: _____

Practice test made from the topics taught in class (Toddle class diary). The AV3 study guide has not been published yet. It is not an official school test.

Read each question carefully. Remember: 1 L = 1,000 mL · 1 kg = 1,000 g · 1 m = 100 cm. In Part F, show your work.

Part A — Multiple choice

3 points

Circle the correct answer.

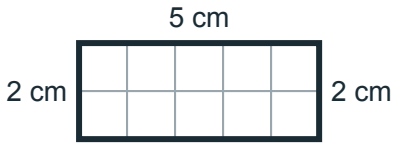
1. Which unit would you use to measure the water in a swimming pool?
   a. milliliters
   b. liters
   c. grams
   d. centimeters

2. Which unit would you use to measure the mass of a paper clip?
   a. grams
   b. kilograms
   c. liters
   d. meters

3. How many milliliters are in **2 liters**?
   a. 20 mL
   b. 200 mL
   c. 2,000 mL
   d. 1,000 mL

4. Which is the **heaviest**?
   a. 500 g
   b. 1 kg
   c. 900 g
   d. 100 g

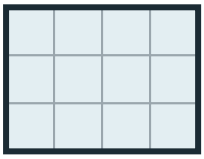
5. What is the **perimeter** of this rectangle?



Rectangle A

a. 7 cm
b. 10 cm
c. 14 cm
d. 12 cm

6. What is the **area** of this rectangle?



Rectangle B

a. 7 squares
b. 12 squares
c. 14 squares
d. 10 squares

Part B — True or false

1.5 points

Write T (true) or F (false). If it is false, write the correct sentence.

7. ( ) 1 kilogram is the same as 1,000 grams.
   Correct it: _____

8. ( ) We measure milk in kilograms.
   Correct it: _____

9. ( ) 1 meter is the same as 10 centimeters.
   Correct it: _____

10. ( ) The area is the space inside a shape.
    Correct it: _____

11. ( ) To measure with a ruler, we start at number 1.
    Correct it: _____

Part C — Complete the sentences

1.5 points

Use the words in the word bank. One word is extra.

Word bank: liters · milliliters · kilograms · grams · centimeters · perimeter

12. A big bottle of juice holds 2 _____ .

13. A spoon of medicine holds 5 _____ .

14. A bag of rice has a mass of 5 _____ .

15. My pencil is 15 _____ long.

16. The distance all the way around a shape is the _____ .

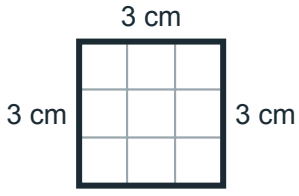
Part D — What are we measuring?

1.2 points

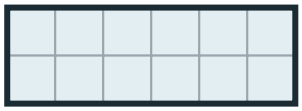
Put an X in the column of the unit you would use.

	liters / mL	kg / g	m / cm
water in a fish tank			
a watermelon			
the height of a door			
a cup of tea			
an apple			
a crayon			

Look at the shapes for Part E



Shape C


Shape D

Shape C: a square with sides of 3 cm. Shape D: count the squares.

Part E — Perimeter and area

1.6 points

Write the answer with the unit.

17. What is the perimeter of Shape C?

_____

18. What is the area of Shape C?

_____

19. What is the area of Shape D?

20. Shape D: each square has sides of 1 cm. What is its perimeter?

Part F — Problems

1.2 points

Show your work. Then write the answer in a complete sentence.

21. Mom has a 1-liter bottle of water. She pours 300 mL into a glass. How many milliliters are left in the bottle?

22. A garden is a rectangle 6 m long and 3 m wide. Leo wants to put a fence all the way around it. How many meters of fence does he need?

Good luck! · Total: 10

Answer key (Gabarito) — Math Practice Test — Measuring detectives

Part A — Multiple choice

1. b) liters
2. a) grams
3. c) 2,000 mL — *1 L = 1,000 mL, so 2 L = 2,000 mL*
4. b) 1 kg — *1 kg = 1,000 g*
5. c) 14 cm — *5 + 2 + 5 + 2 = 14*
6. b) 12 squares — *3 rows of 4 = 12*

Part B — True or false

7. T (true)
8. F (false) — We measure milk in liters (or milliliters).
9. F (false) — 1 meter is the same as 100 centimeters.
10. T (true)
11. F (false) — To measure with a ruler, we start at zero.

Part C — Complete the sentences

12. liters
13. milliliters
14. kilograms
15. centimeters
16. perimeter

Part D — What are we measuring?

water in a fish tank: liters / mL
a watermelon: kg / g
the height of a door: m / cm
a cup of tea: liters / mL
an apple: kg / g
a crayon: m / cm

Part E — Perimeter and area

17. 12 cm — *3 + 3 + 3 + 3*
18. 9 squares — *3 rows of 3*
19. 12 squares — *2 rows of 6*
20. 16 cm — *6 + 2 + 6 + 2*

Part F — Problems

21. **Expected answer (example):** There are 700 mL left in the bottle (1 L = 1,000 mL; 1,000 − 300 = 700).
    Para o responsável, confira: troca 1 L por 1.000 mL; conta certa (700 mL); resposta em frase (0,6)
22. **Expected answer (example):** He needs 18 meters of fence (6 + 3 + 6 + 3 = 18).

Para o responsável, confira: entende que "all the way around" é o perímetro; soma os 4 lados (18 m); resposta em frase (0,6)